VSB TECHNICAL UNIVERSITY OF OSTRAVA | FACULTY OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

Bachelor Thesis Assignment

Student: **Dominik Chrascina**

Study Programme: B0613A140014 Computer Science

Title: Software Synthesizer with Integration into the JUCE Framework
Softwarový syntezátor s integrací do frameworku JUCE

The thesis language: English

Description:

A software synthesizer is an electronic musical instrument that creates sound by generating a digital audio signal using computational methods. The aim of this thesis is to design and implement a multiplatform software synthesizer using PortAudio and libremidi libraries with integration into JUCE framework.

The software synthesizer will be supported by current DAW applications (such as FL Studio and Ableton Live) in the form of an audio plugin. The synthesizer will support playing through both a computer keyboard and connected MIDI devices, support polyphony, and allow users to modify the generated sound using built-in effects, ADSR parameters, and waveform selection.

The objectives of the thesis are:
1. To design and implement a multiplatform synthesizer core independent of any specific plugin framework. The use of C++ is recommended for the implementation.
2. To integrate the developed core into the JUCE framework and create a functional audio plugin in a format supported by current DAW applications (e.g., VST3).
3. To verify the functionality of the resulting plugin in commonly used DAW applications.

References:

1. JUCE Framework Documentation. Available from: https://juce.com/learn/documentation
2. PortAudio Tutorials. Available from: https://portaudio.com/docs.html
3. libremidi Documentation. Available from: https://celtera.github.io/libremidi/
4. Envelope generators. Available from: https://www.earlevel.com/main/2013/06/01/envelope-generators/
5. Audio EQ Cookbook. Available from: https://www.w3.org/TR/audio-eq-cookbook/

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Extent and terms of a thesis are specified in directions for its elaboration that are opened to the public on the web sites of the faculty.

Supervisor: **Ing. Tomáš Luber, Ph.D.**

Date of issue:

Date of submission:

Study programme guarantor: Ing. Tomáš Fabián, Ph.D.

In IS EDISON assigned:

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